

Long LM vs ANS Schedule Sweep

GIANT v3 Long pipeline sweep comparing LM-heavy and answer-heavy schedules on the same diversified natural-record curriculum.

Setup

Model 38.64M params 448 dim, 7 heads, 12 layers, FF 1792	Curriculum 64 to 128 to 256 to 512 single 512-context model, staged seq lengths	Eval set 128 held-out ctx512 samples exact-match on final answer token
Diversity update enabled richer alias and filler record templates before the sweep	GPU 1x RTX A6000 RunPod secure cloud	Checkpoint cadence every 500 steps 24 evaluation points per run

Compared schedules

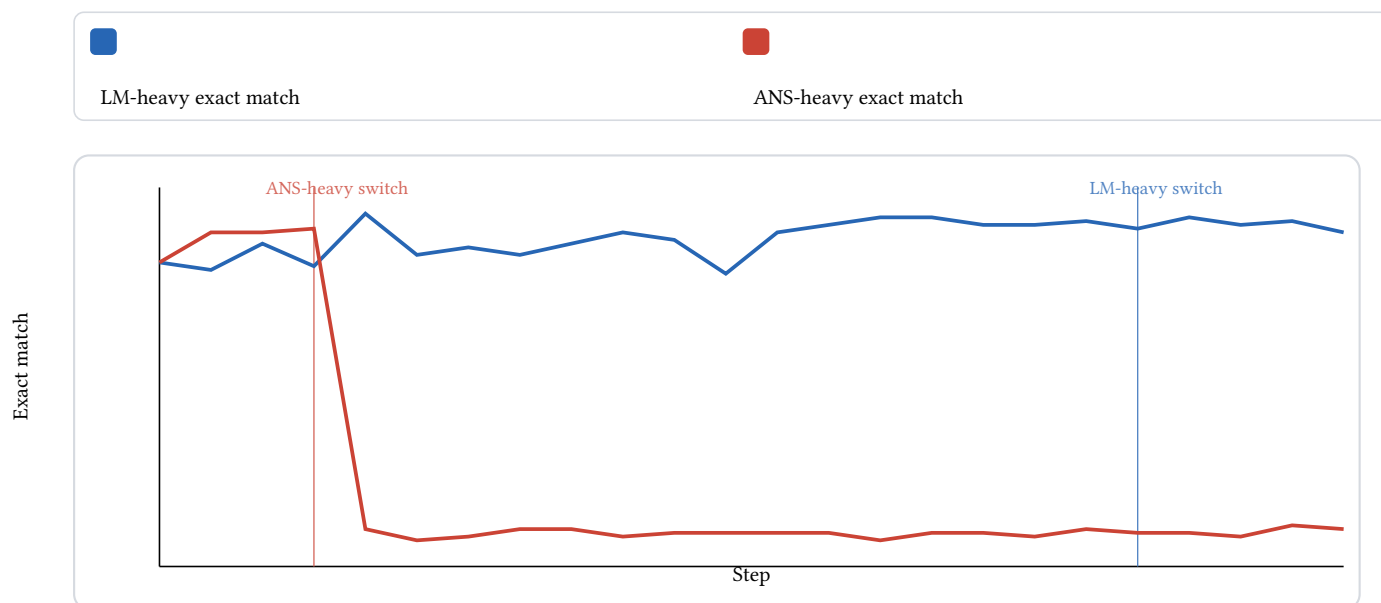
Both runs use the same datasets, model, learning rate, and total 12,000 steps. Only the split between LM and answer-only supervision changes.

- LM-heavy: 10,000 LM steps then 2,000 answer-only steps
- ANS-heavy: 2,000 LM steps then 10,000 answer-only steps

The phase boundaries are marked in the charts:

- step 2,000: ANS-heavy switches from LM warmup into answer-heavy training
- step 10,000: LM-heavy switches from LM-heavy training into answer-only finishing

Held-out exact match over training

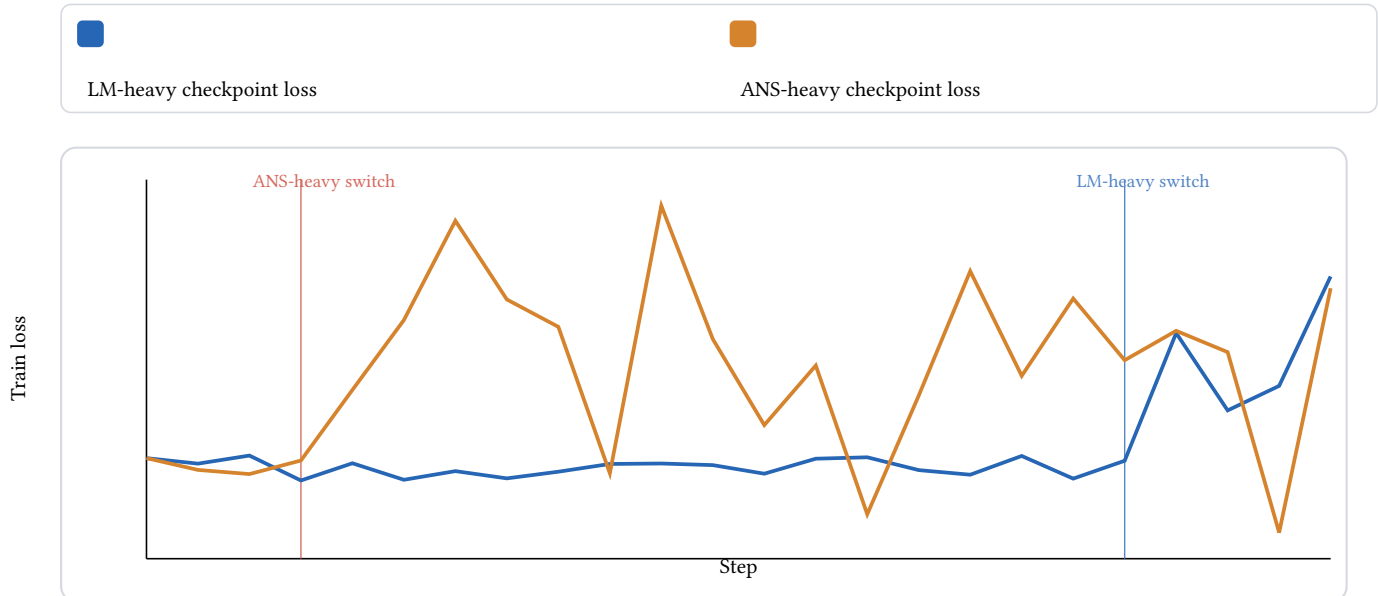


The result is decisive.

- LM-heavy stays strong through the full run and peaks at $87 / 128 = 67.97\%$ at step 2,500

- LM-heavy finishes at $82 / 128 = 64.06\%$
- ANS-heavy is competitive only during the initial LM warmup and peaks at $83 / 128 = 64.84\%$ at step 2,000
- Once ANS-heavy enters answer-dominated training, held-out exact match collapses to near zero and never recovers

Checkpoint train loss



This chart explains why the answer-heavy run is misleading if we only watch training loss.

- ANS-heavy often reports low or unstable checkpoint loss inside the answer-only stages
- but those low losses do **not** translate into held-out retrieval quality
- the run is overfitting the final supervised position and forgetting the broader retrieval behavior needed at `ctx512`

Main finding

For this LongGIANT natural-record setup, more answer-only supervision is **not** better. The model needs a long LM-heavy curriculum to keep the document structure, alias resolution, and retrieval cues stable across the long context.

The strongest simple interpretation is:

- the first 2,000 LM steps already build a viable retrieval representation
- pushing too hard on answer-only after that destroys generalization
- a small finishing block of answer-only steps helps, but it should come late and stay limited

Concrete results

Run	Best step	Best EM	Final EM
LM-heavy	2500	$87 / 128 = 67.97\%$	$82 / 128 = 64.06\%$
ANS-heavy	2000	$83 / 128 = 64.84\%$	$3 / 128 = 2.34\%$

Practical takeaway for the next Long runs

- keep the diversified natural-record renderer
- keep the context ladder 64 -> 128 -> 256 -> 512
- stay LM-heavy for most of training
- if answer-only is used, keep it short and late
- do **not** launch another long answer-heavy sweep unless the objective itself changes

Artifact locations

- configs:
 - GIANT/v3/Configs/Training/Long/long_40m_l1_sweep_lmheavy.yml
 - GIANT/v3/Configs/Training/Long/long_40m_l1_sweep_ansheavy.yml
- local logs:
 - /proj/giant-data/GIANT/single-gpu/checkpoints/long/lmheavy_10k_lm_2k_ans/40m/l1_ctx512/logs.txt
 - /proj/giant-data/GIANT/single-gpu/checkpoints/long/ansheavy_2k_lm_10k_ans/40m/l1_ctx512/logs.txt
- local key checkpoints:
 - /proj/giant-data/GIANT/single-gpu/checkpoints/long/lmheavy_10k_lm_2k_ans/40m/l1_ctx512/step_0002500.npz
 - /proj/giant-data/GIANT/single-gpu/checkpoints/long/lmheavy_10k_lm_2k_ans/40m/l1_ctx512/step_0012000.npz
 - /proj/giant-data/GIANT/single-gpu/checkpoints/long/ansheavy_2k_lm_10k_ans/40m/l1_ctx512/step_0002000.npz
 - /proj/giant-data/GIANT/single-gpu/checkpoints/long/ansheavy_2k_lm_10k_ans/40m/l1_ctx512/step_0012000.npz

The sweep answers the scheduling question clearly: for this benchmark, LM-heavy wins by a large margin once long-context generalization matters.